

MTH166

Lecture-25

Solution of Laplace Equation

Topic:

Solution of Partial Differential Equations

Learning Outcomes:

To solve two dimensional Laplace Equation

Problem. Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ [2D-Laplace Equation]

Solution. The given two dimensional Laplace equation is:

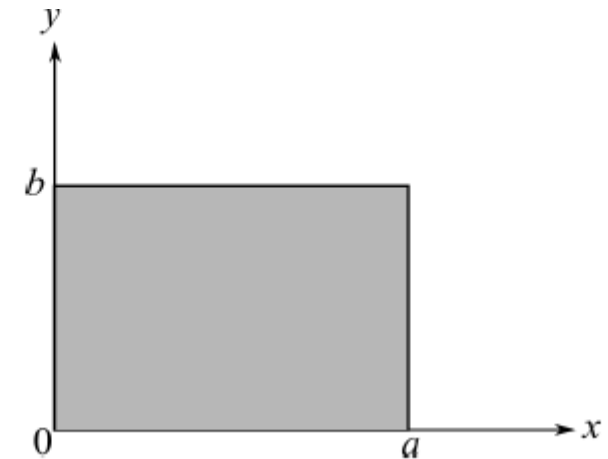
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad (1) \quad 0 \leq x \leq a, 0 \leq y \leq b$$

$$\text{Let solution be: } u(x, y) = XY \quad (2) \quad \text{where } X = f(x), Y = g(y)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = X''Y \quad \text{and} \quad \frac{\partial^2 u}{\partial y^2} = XY''$$

$$\text{Equation (1) becomes: } X''Y + XY'' = 0$$

$$\Rightarrow \frac{X''}{X} = -\frac{Y''}{Y} = k \quad (\text{Say})$$



As k can take three values: zero, positive or negative, so we have following three cases.

Case 1. When $k = 0$

$$\frac{X''}{X} = k \quad \Rightarrow \frac{X''}{X} = 0 \quad \Rightarrow X'' = 0 \quad \Rightarrow X = ax + b$$

$$\text{Also } -\frac{Y''}{Y} = k \quad \Rightarrow -\frac{Y''}{Y} = 0 \quad \Rightarrow Y'' = 0 \quad \Rightarrow Y = cy + d$$

Required solution of equation (1) is:

$$u(x, y) = XY = (ax + b)(cy + d)$$

Case 2. When $k = p^2$ (Positive)

$$\frac{X''}{X} = k \quad \Rightarrow \frac{X''}{X} = p^2 \quad \Rightarrow X'' - p^2 X = 0$$

$$\text{S.F. } (D^2 - p^2)X = 0$$

$$\text{A.E. } (D^2 - p^2) = 0 \quad \Rightarrow D = \pm p$$

$$\therefore X = ae^{px} + be^{-px}$$

$$\text{Also } -\frac{Y''}{Y} = k \Rightarrow -\frac{Y''}{Y} = p^2 \Rightarrow Y'' + p^2Y = 0$$

$$\underline{\text{S.F.}} (D^2 + p^2)Y = 0$$

$$\underline{\text{A.E.}} (D^2 + p^2) = 0 \Rightarrow D = \pm ip$$

$$\therefore Y = c \cos py + d \sin py$$

Required solution of equation (1) is:

$$u(x, y) = XY = (ae^{px} + be^{-px})(c \cos py + d \sin py)$$

Case 3. When $k = -p^2$ (Negative)

$$\frac{X''}{X} = k \Rightarrow \frac{X''}{X} = -p^2 \Rightarrow X'' + p^2X = 0$$

$$\underline{\text{S.F.}} (D^2 + p^2)X = 0$$

$$\underline{\text{A.E.}} (D^2 + p^2) = 0 \quad \Rightarrow D = \pm ip$$

$$\therefore X = a \cos px + b \sin px$$

$$\text{Also } -\frac{Y''}{Y} = k \quad \Rightarrow -\frac{Y''}{Y} = -p^2 \quad \Rightarrow Y'' - p^2 Y = 0$$

$$\underline{\text{S.F.}} (D^2 - p^2)Y = 0$$

$$\underline{\text{A.E.}} (D^2 - p^2) = 0 \text{ Type equation here. } \Rightarrow D = \pm p$$

$$\therefore Y = ce^{py} + de^{-py}$$

Required solution of equation (1) is:

$$u(x, y) = XY = (a \cos px + b \sin px) (ce^{py} + de^{-py})$$

Note: Laplace Equation

Equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

Nature: Elliptic

Solution: 1. $u(x, y) = (ax + b)(cy + d)$

2. $u(x, y) = (ae^{px} + be^{-px})(c \cos py + d \sin py)$

or

$$u(x, y) = (a \cosh px + b \sinh px)(c \cos py + d \sin py)$$

3. $u(x, y) = (a \cos px + b \sin px)(ce^{py} + de^{-py})$

or

$$u(x, y) = (a \cos px + b \sin px)(c \cosh py + d \sinh py)$$

Problem. Prove that a two dimensional heat equation becomes Laplace equation in steady state.

Solution. The two dimensional heat equation is:

$$\frac{\partial u}{\partial t} = C^2 \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right] \quad (1)$$

In steady-state: $\frac{\partial u}{\partial t} = 0$

So, in steady-state equation (1) becomes:

$$0 = C^2 \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right] \Rightarrow \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right] = 0 \quad (\text{As } C^2 \neq 0)$$

which is a two dimensional Laplace Equation.

